

Overview of NoSQL

Sure! Let's dive into the concept of NoSQL databases in simple terms.

NoSQL databases are a type of database that doesn't use the traditional table format of rows and columns like you might find in a typical spreadsheet. Instead, they allow for more flexible ways to store and manage data. Think of it like a toolbox: while a regular database is like a neatly organized toolbox with specific compartments for each tool, a NoSQL database is more like a big, open toolbox where you can throw in all sorts of tools without worrying about fitting them into specific slots. This flexibility makes NoSQL databases great for handling different types of data, especially when that data is constantly changing.

For example, imagine you have a social media app. You might need to store user profiles, posts, comments, and likes. Each of these pieces of information can be quite different from one another. A NoSQL database allows you to store all of this information in a way that makes sense for your app, without needing to stick to a strict format. This means you can adapt and grow your app more easily as new features are added or as user needs change.

What is NoSQL?

- The NoSQL name was introduced during an open-source event on distributed databases
- NoSQL doesn't mean 'No SQL'
NoSQL means 'Not only SQL'

NoSQL
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'Not only
SQL'

- First, let's talk about the name NoSQL. The name NoSQL was introduced at an event to discuss all the new open source distributed databases, and the name has stuck ever since. Contrary to what it sounds like, NoSQL stands for "not only SQL" and not "no SQL".

What is NoSQL?

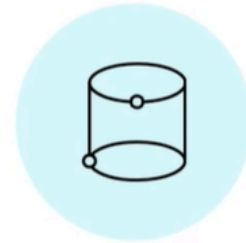
- Refers to family of databases that vary widely in style and technology
- However, they share common traits:
 - Non-relational
 - Not standard row and column type RDBMS
- Could be referred to as 'Non-relational databases'

- The name NoSQL really describes what it is not. That is, it refers to a family of databases that vary widely in style and technology. But which all share a common trait in that they are non relational in nature. Meaning they are not a standard row and column relational database management system or RDBMS. Therefore, a better name to describe these databases would be non relational

What is NoSQL?

NoSQL databases:

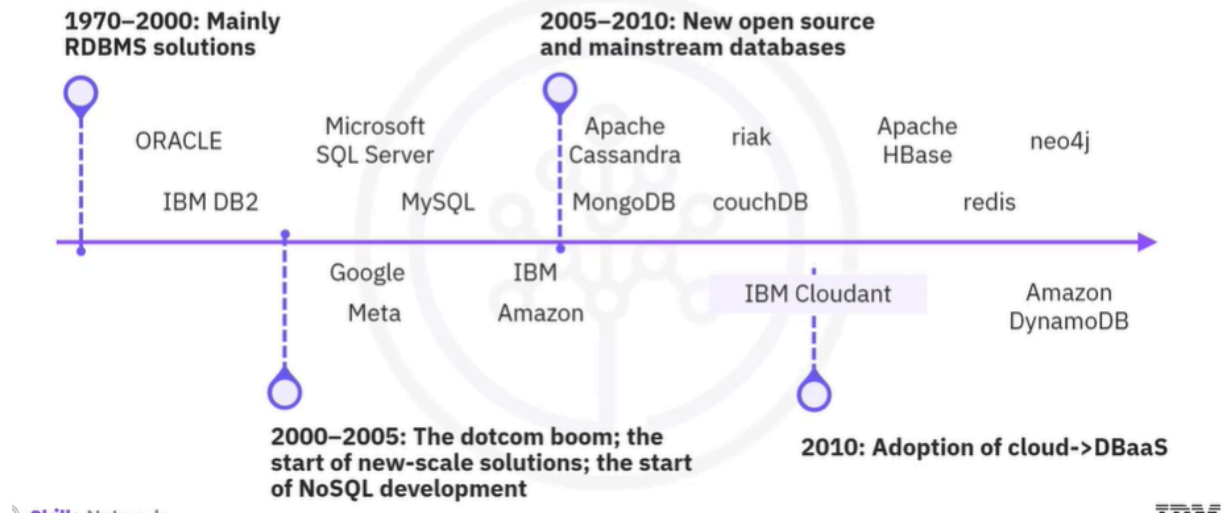
- Provide new ways of storing and querying data
- Address several issues for modern apps
- Provide a flexible schema
- Scale to meet demand
- Are distributed systems



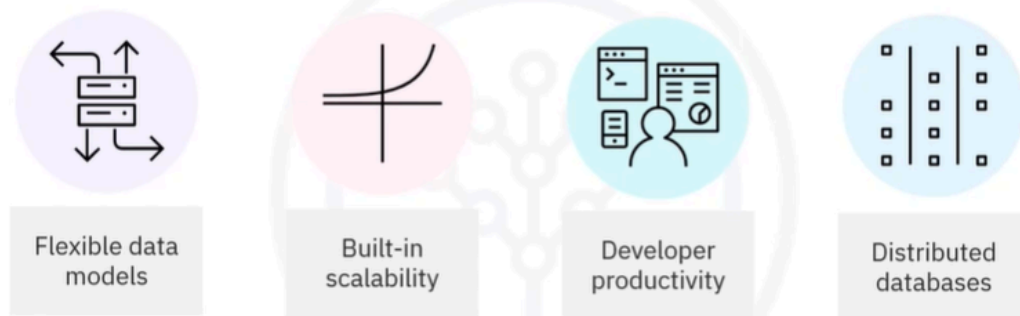
NoSQL
Databases

- NoSQL databases provide new ways of storing and querying data that address many of the issues that modern applications experience. Most importantly, unlike relational databases, NoSQL databases don't require fixed schemas, making them suitable for evolving use cases. NoSQL databases horizontally scale easily, allowing you to add more capacity for data and traffic as the demand grows. NoSQL databases are distributed systems that also provide native fault tolerance and availability.

History of NoSQL



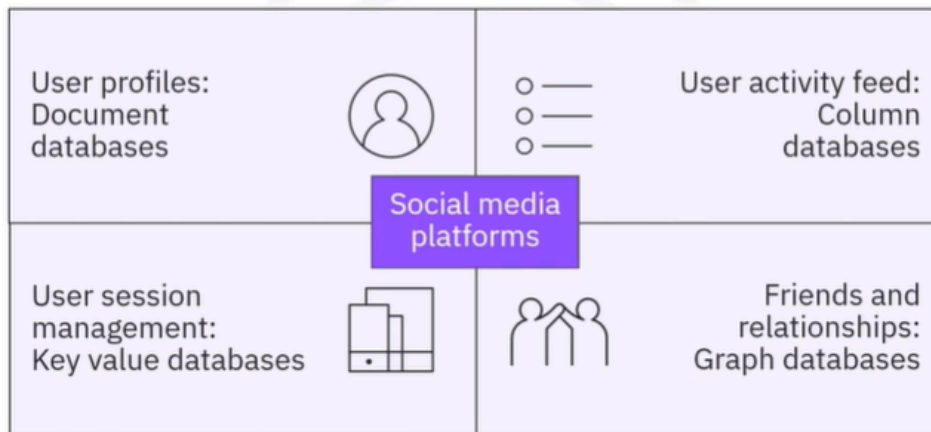
Why NoSQL?



- Let's take a closer look at why NoSQL. NoSQL capabilities support a flexible data model, which means you can store unstructured or semi structured data more easily. Some NoSQL databases provide native, built in horizontal and vertical scaling capabilities. Developers can work faster and more productively with data structures that match the application's needs for reads and writes. NoSQL databases work in distributed environments and provide high availability and fault tolerance.

NoSQL in use: Social media example

Businesses can use a mix of NoSQL databases



- Businesses often use a mix of NoSQL databases for their needs. Let's take the example of social media platforms. Social media platforms use document databases to manage user profiles. Column databases store user activity feed information, while key value databases help organizations manage user sessions and speed user access to frequently access data. And graph databases do the essential work of keeping people connected by storing information about friends and relationships.

Recap

In this video, you learned that:

- The name 'NoSQL' stands for **Not only **SQL****
- NoSQL refers to a class of databases that are non-relational in architecture
- Implementations of NoSQL databases differ technically but share common traits
- NoSQL databases are distributed systems that easily horizontally scale, and provide native fault tolerance and high availability